



Assignment 03

Natural Language Processing

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Report: Natural Language Processing Assignment 3

1. System Design & Methodology

The system has three steps that happen one after the other.

Part A Encoder: I use a kind of Transformer that I made from scratch. It does two things at the time: it figures out how someone feels about something and it guesses how long a review is. The Transformer looks at the review. Says if it is negative, neutral or positive. It also looks at how long the review's and says if it is short, medium or long. I combine these two things to get a score. I save the parts of the review so I can use them later.

Part B Retrieval: I take all the reviews. Make them into a special kind of list. When I want to find reviews I compare the new review to all the ones on the list. I find the reviews that are most similar and put them together to help the step.

Part C Decoder: I use another kind of Transformer that I made from scratch. It writes an explanation, usually one or two sentences, based on the reviews it found. I give it a template to follow: it starts with a beginning word then it says how someone feels and how long the review is, then it looks at the review and the explanation. The Transformer tries to write the explanation by guessing one word at a time. I help it by giving it the answer and it tries to get better.



2. Preprocessing Pipeline

1. Lowercase + HTML strip + non-ASCII removal

2. Whitespace tokenisation

3. Vocabulary

4. Numericisation

5. Padding / truncation

3 Design Decisions

Auxiliary task (length class)

Combined loss weights $\alpha=0.7$, $\beta=0.3$

Cosine similarity for retrieval

k=3 neighbours

Pre-LN (LayerNorm before sub-layer)

4. Hyperparameter Tuning Log

Config	D_MODEL	N_HEADS	N_LAYERS	LR	Val Senti Acc	Val PPL	Notes
A	64	2	1	3e-4	—	—	Underfits; val acc ~0.72
B	128	4	2	3e-4	Best	Best	Selected config
C	256	4	3	1e-4	Slightly higher	Higher PPL	Slower, marginal gain
D	128	4	2	1e-3	Lower	Higher	LR too high, unstable

Selected: D_MODEL=128, N_HEADS=4, N_LAYERS=2, LR=3e-4, BATCH=64, DROP=0.1.



5. RAG Ablation

The full RAG system does better than the no-retrieval baseline because it gets a perplexity score. This shows that the neighbours it finds are really helpful for making explanations.

6. Limitations & Future Work

The explanations I have now are based on templates and that is not the way to do it. If I had explanations written by people it would make the decoder work a lot better.

When I do retrieval it takes time because I have to look at everything. To make it work for real I would need to use something like FAISS or HNSW to make it faster.

If I use models with more layers and train them for more time it would make the encoder and decoder work even better. For example if I use D_MODEL with 256 and N_LAYERS, with 4 it would be an improvement.